Rajat Ramesh Dungarwal

AI/ML Engineer

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Professional Summary

AI/ML Engineer with 3+ years of experience designing, developing, and deploying scalable machine learning and deep learning solutions across cloud environments. Expertise in Python, NLP, Computer Vision, and MLOps pipelines, with strong experience in handling large-scale datasets and delivering business-driven AI solutions. Proven track record of improving model performance, optimizing data pipelines, and deploying production-ready models with measurable impact on business outcomes.

Professional Experience

AI/ML Engineer | IBM | USA **June 2025 – Present**

- Designed and deployed end-to-end machine learning models using Python, TensorFlow, and PyTorch to solve large-scale business problems, processing over 15M+ records, improving prediction accuracy by 28% and enabling data-driven decision-making across enterprise applications.
- Built scalable data pipelines using PySpark, Apache Spark, and Kafka to process real-time streaming data, reducing data latency by 35% and enabling faster model training and inference workflows in production environments.
- Implemented advanced NLP models using Hugging Face Transformers, BERT, and SpaCy to extract insights from unstructured data, improving text classification accuracy by 30% and automating document processing workflows.
- Deployed machine learning models on AWS SageMaker and Azure ML using Docker and Kubernetes, ensuring high availability and scalability, achieving 99.8% uptime and reducing deployment time by 40%.
- Developed recommendation systems and predictive analytics solutions using Scikit-learn and XGBoost, increasing customer engagement metrics by 22% and improving personalization across digital platforms.
- Built MLOps pipelines using MLflow, Airflow, and CI/CD tools, automating model lifecycle management and reducing manual intervention by 45%, ensuring faster model retraining and deployment cycles.
- Performed hyperparameter tuning and model optimization using Grid Search and Bayesian optimization, improving model performance by 18% and reducing overfitting across multiple ML use cases.
- Collaborated with cross-functional teams including data engineers and product managers using Jira and Confluence, delivering AI solutions aligned with business goals and reducing project delivery timelines by 20%.
- Implemented model monitoring and drift detection systems to ensure continuous model performance, reducing model degradation issues by 25% and improving long-term reliability of deployed solutions.

Machine Learning Engineer | Wipro | India **July 2020 – July 2023**

- Developed and deployed machine learning models using Python, Scikit-learn, and TensorFlow on structured and unstructured datasets, improving predictive accuracy by 25% and supporting business analytics use cases across multiple domains.
- Built ETL pipelines using SQL, Python, and Apache Spark to process 10M+ records, improving data quality and reducing processing time by 30%, enabling efficient feature engineering and model training.
- Designed and implemented deep learning models for image and text data using CNNs and RNNs, improving classification performance by 27% and enabling automation of data-driven workflows.
- Worked on NLP use cases using NLTK, SpaCy, and Transformers, improving sentiment analysis accuracy by 32% and enhancing customer feedback analysis systems.
- Deployed models into production using Docker and Kubernetes, reducing deployment errors by 35% and improving scalability across cloud-based applications.
- Leveraged AWS (S3, EC2, Lambda) for model hosting and data processing, reducing infrastructure costs by 20% while improving system scalability and performance.
- Implemented data visualization dashboards using Tableau and Power BI, enabling stakeholders to track model performance and improving reporting efficiency by 40%.
- Conducted A/B testing and performance evaluation of ML models, improving business decision-making accuracy and increasing model adoption across teams by 18%.
- Collaborated with cross-functional teams in Agile environments, using Git, Jira, and Confluence, ensuring smooth project delivery and improving team productivity by 22%.

Technical Skills

- **Programming & Scripting:** Python, R, SQL, Scala, Bash
- **Machine Learning Frameworks:** Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM
- **Deep Learning & NLP:** Transformers, Hugging Face, BERT, GPT, SpaCy, NLTK
- **Big Data & Processing:** Apache Spark, PySpark, Hadoop, Kafka, Databricks
- **Cloud Platforms:** AWS (SageMaker, S3, Lambda), Azure ML, GCP AI Platform
- **MLOps & Deployment:** Docker, Kubernetes, MLflow, CI/CD, Airflow, Kubeflow
- **Data Engineering:** ETL Pipelines, Data Warehousing, Snowflake, Redshift, BigQuery
- **Visualization & BI:** Tableau, Power BI, Matplotlib, Seaborn
- **Databases:** MySQL, PostgreSQL, MongoDB, Cassandra
- **Version Control & Tools:** Git, GitHub, GitLab, Jira, Confluence
- **Model Monitoring & Optimization:** A/B Testing, Model Drift Detection, Hyperparameter Tuning
- **AI Techniques:** Predictive Modeling, Classification, Regression, Time Series, Recommendation Systems

Projects

AI Analytics Copilot (Natural Language to SQL Insights Platform)

Python, FastAPI, PostgreSQL, Streamlit, Ollama (Local LLM), SQL

- Developed an AI analytics platform that converts natural language queries into SQL using a local large language model, enabling faster, more accurate data analysis across structured datasets.

Monthly Retail Sales Dashboard (E-Commerce Business Intelligence)

Power BI, Python, SQL, Excel, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

- Built an analytics pipeline using Python, SQL, and Power BI to transform 10K+ transactional records into KPI-driven dashboards for monitoring trends and performance metrics.

Thing Translator (AI Product and Research Project)
TensorFlow, Android, Java, NLP, TTS, Image Classification

- Engineered a real-time image classification pipeline using TensorFlow Lite, compressing model size by 45% and reducing average inference time to 0.8 seconds on Android devices.

Personal Finance Tracker (Financial Planning Application)
Flask, MySQL, Git, Docker, REST APIs, OAuth 2.0, JWT

- Constructed a complete CI/CD pipeline with automated testing and deployment using Jenkins and Docker, enabling a 90% code coverage and 30% faster release of new features

Education

Master of Science in Computer Science GPA: 3.92/4.0 **Aug 2023– May 2025**
University of Texas at Arlington, TX, USA
Bachelor's in Computer Engineering GPA: 3.6/4.0 **Aug 2017 – June 2021**
University of Mumbai, India

Certification

Google - Machine Learning Operations (MLOps) for Generative AI
IBM - Data Science
Authored & Published - Thing Translator (Eliva Press; available on Amazon) contributing applied AI research in ML & NLP
Prediction using Machine Learning Algorithms